Harold Soh Soon Hong

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Research Interests: Trustworthy Embodied AI, Human-Robot Interaction, Robotics, Machine Learning and Artificial Intelligence.

Professional Experience

National University of Singapore, Singapore.
- Associate Professor, Dept. of Computer Science — **Jan 2025-Present**
- Assistant Professor, Dept. of Computer Science — **Feb 2017-Dec 2024**
- Director, Computing Horizons Office — **Nov 2025-Present**
- PI, Smart Systems Institute — **Jan 2022-Present**
- Associate Director, NUS AI Lab — **Jan 2023-Mar 2024**

University of Toronto, Toronto, ON, Canada.
- Postdoctoral Research Fellow, Cognitive Engineering Lab & Data-Driven Decision Making Lab (D3M) — **Oct 2015-Dec 2016**

Singapore-MIT Alliance for Research & Technology (Smart), Singapore.
- SMART Postdoctoral Scholar, Future Urban Mobility — **Oct 2013-Sept 2015**

The Institute of High Performance Computing, Agency for Science, Technology and Research (A*STAR), Singapore
- Senior Research Officer, Advanced Computing — **Apr 2008-Jul 2009**
- Research Officer, Advanced Computing — **Feb 2006-Mar 2008**

Academic Qualifications

Imperial College London, London, United Kingdom
Ph.D., **Artificial Intelligence & Robotics**, November 2013
- Research Topic: Online Spatio-Temporal Learning and Prediction for Robotic Systems [**Nominated for Eryl Cadwaladr Davis Prize 2013/2014**]
- Advisor: Prof. Yiannis Demiris

The University of Melbourne, Victoria, Australia.
Masters (S.S.E.), March 2006.

The University of California, Davis, California, USA.
Bach. Arts & Sciences (Dual Major), Computer Science, Economics, June 2004.
University of California Regents Scholar

Awards and Achievements

NRF Investigatorship, National Research Foundation, Singapore, 2026.
NUS University Teaching Awards Honour Roll, 2025–2029.
Invited Keynote, IEEE ICRA, 2024 and 2025.
Early Career Spotlight Award, Robotics: Science and Systems (R:SS), 2023.
Best Paper Award, IEEE Trans. Affec. Computing (T-AFFC), 2022.
Best Paper Award, IEEE/RSJ IROS'21.
NUS Annual Teaching Excellence Award (ATEA), 2021/22/24
Faculty Teaching Excellence Award (FTEA), 2018/19/20/21
Best Paper Award Finalist, RSS'18.
Best Paper Award Finalist, ACM/IEEE HRI'18.
Best Paper Award Finalist, ACM RecSys'18.
Smart Scholar Postdoctoral Fellowship Award, 2013.
James Dyson Award National Finalist (UK), 2012.
Best Cognitive Robotics Paper Award Finalist, IEEE/RSJ IROS'12.
The Design London Fellowship, 2012.
The Khazanah Global Scholarship, 2009-2013.
IEEE CEC Best Presentation Award, 2006

Peer-Reviewed Publications

The top venues in Computer Science are conferences. For robotics, the top conferences are ICRA, IROS, RSS, and HRI. The best general conferences in AI/ML are NeurIPS, ICLR, ICML, AAAI, and IJCAI.

Jiang, P., Wang, M., Lin, K., Hao, C., & Soh, H. (2026). Guided Streaming Stochastic Interpolant Policy. In *Robotics: Science and Systems (RSS)*. Retrieved from https://arxiv.org/abs/2605.10051

Zhai, X., Huang, Z., Wu, L., Zhao, Q., Yu, Q., Ren, J., ... Soh, H. (2026). SkillVLA: Tackling Combinatorial Diversity in Dual-Arm Manipulation via Skill Reuse. In *Robotics: Science and Systems (RSS)*. Retrieved from https://arxiv.org/abs/2603.03836

Yu, X., Cai, F., Wang, M., Fan, X., & Soh, H. (2026). Conflict-Aware Additive Guidance for Flow Models under Compositional Rewards. In *International Conference on Machine Learning (ICML)*. Retrieved from https://arxiv.org/abs/2605.20758

Bi, J., Ma, K. Y., Hao, C., Shou, M. Z., & Soh, H. (2026). VLA-Touch: Enhancing Vision-Language-Action Models with Dual-Level Tactile Feedback. *IEEE Robotics and Automation Letters*, *11*(7), 8487–8494. Retrieved from https://arxiv.org/abs/2507.17294 doi: 10.1109/LRA.2026.3692345

Wang, J., Chen, J., Liu, D., Da, J., Hu, J., Xue, Z., ... Soh, H. (2026). TOPO-Bench: An Open-Source Topological Mapping Evaluation Framework with Quantifiable Perceptual Aliasing. In *IEEE International Conference on Robotics and Automation (ICRA)*. Retrieved from https://arxiv.org/abs/2510.04100

Hao, C., Zhai, X., Liu, Y., & Soh, H. (2026). Abstracting Robot Manipulation Skills via Mixture-of-Experts Diffusion Policies. In *International Conference on Learning Representations (ICLR)*. Retrieved from https://arxiv.org/abs/2601.21251

Heng, A., & Soh, H. (2026). Know When to Abstain: Optimal Selective Classification with Likelihood Ratios. In *International Conference on Learning Representations (ICLR)*. Retrieved from https://arxiv.org/abs/2505.15008

Feng, Z., Yin, H., Ong, Y.-S., & Soh, H. (2026). Masked Skill Token Training for Hierarchical Off-Dynamics Transfer. In *International Conference on Learning Representations (ICLR)*. Retrieved from https://openreview.net/forum?id=K4ngUOra9m

Sabbah, M., Wojciechowski, K., Soh, H., Hsu, D., Righetti, L., Mansard, N., ... Bonnet, V. (2025). Optimal Motion Prediction for Human-to-Robot Handovers. In *IEEE International Conference on Robotics and Biomimetics (ROBIO)*. Retrieved from https://hal.science/hal-04970974

Liang, X., Liu, Z., Lin, K., Gu, E., Ye, R., Nguyen, T., ... Bhattacharjee, T. (2025). OpenRoboCare: A Multimodal Multi-Task Expert Demonstration Dataset for Robot Caregiving. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Retrieved from https://emprise.cs.cornell.edu/robo-care/

Hao, C., Xiao, A., Xue, Z., & Soh, H. (2025). CHD: Coupled Hierarchical Diffusion for Long-Horizon Tasks. In *Conference on Robot Learning (CoRL)*. Retrieved from https://arxiv.org/abs/2505.07261

Hao, C., Lin, K., Xue, Z., Luo, S., & Soh, H. (2025). DISCO: Language-Guided Manipulation with Diffusion Policies and Constrained Inpainting. *IEEE Robotics and Automation Letters (RA-L)*, *10*(10), 9726-9733.

Wang, J., Liu, D., Chen, J., Da, J., Qian, N., Man, T. M., & Soh, H. (2025). GeNIE: A Generalizable Navigation System for In-the-Wild Environments. *IEEE Robotics and Automation Letters*. Retrieved from https://arxiv.org/abs/2506.17960

Yu, S., Lin, K., & Soh, H. (2025). Demonstrating the Octopi-1.5 Visual-Tactile-Language Model. In *Robotics: Science and Systems (RSS)*. Retrieved from https://arxiv.org/abs/2507.09985

Kästner, L., Shcherbyna, V., Soh, H., Truong, G. N. H., Anh, D. D., Kien, T. M., ... Schreff, M. H.-K. (2025). Demonstrating Arena 5.0: A Photorealistic ROS2 Simulation Framework for Developing and Benchmarking Social Navigation. In *Robotics: Science and Systems (RSS)*. Retrieved from https://www.roboticsproceedings.org/rss21/p092.html doi: 10.15607/RSS.2025.XXI.092

Feng, Z., Luan, H., Ma, K. Y., & Soh, H. (2025). Diffusion Meets Options: Hierarchical Generative Skill Composition for Temporally-Extended Tasks. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Bi, J., Lim, K., Chen, K., Huang, Y., & Soh, H. (2025). Imitation Learning with Limited Actions via Diffusion Planners and Deep Koopman Controllers. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Shcherbyna, V., Kästner, L., Diaz, D., Nguyen, H. G., Schreff, M. H.-K., Lenz, T., ... Soh, H. (2025). Arena 4.0: A Comprehensive ROS2 Development and Benchmarking Platform for Human-centric Navigation Using Generative-Model-based Environment Generation. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Lim, E., Tan, V., & Soh, H. (2024). Stochastic bandits for egalitarian assignment. *Transactions on Machine Learning Research*. Retrieved from `https://openreview.net/forum?id=kmKVJl2JWo`

Heng, A., Ansari, A. F., & Soh, H. (2024). Generative modeling with flow-guided density ratio learning. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (pp. 250–267).

Heng, A., Thiery, A. H., & Soh, H. (2024). Out-of-distribution detection with a single unconditional diffusion model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Zhang, B., & Soh, H. (2024). Extract, Define, Canonicalize: An LLM-based Framework for Knowledge Graph Construction. In *Empirical Methods in Natural Language Processing (EMNLP)*.

Feng, Z., Luan, H., Goyal, P., & Soh, H. (2024). LTLDoG: Satisfying Temporally-Extended Symbolic Constraints for Safe Diffusion-Based Planning. *IEEE Robotics and Automation Letters (RA-L)*, *9*(10), 8571 - 8578.

Yu, S., Lin, K., Xiao, A., Duan, J., & Soh, H. (2024). Octopi: Object property reasoning with large tactile-language models. In *Robotics: Science and Systems (R:SS)*.

Taunyazov, T., Lin, K., & Soh, H. (2024). Grace: Balancing multiple criteria to achieve stable, collision-free, and functional grasps. In *Robotics: Science and Systems (R:SS)*.

Chen, K., Lim, E., Lin, K., Chen, Y., & Soh, H. (2024). Behavioral refinement via interpolant-based policy diffusion. In *Robotics: Science and Systems (R:SS)*.

Kästner, L., Shcherbyna, V., Zeng, H., Le, T. A., Schreff, M. H.-K., Osmaev, H., ... Lambrecht, J. (2024). Demonstrating Arena 3.0: Advancing Social Navigation in Collaborative and Highly Dynamic Environments. In *Proceedings of Robotics: Science and Systems (R:SS)*.

Heng, A., & Soh, H. (2023). Selective amnesia: A continual learning approach to forgetting in deep generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*. [**Spotlight, < 3.1% of submissions**]

Hu, S., Soh, H., & Piliouras, G. (2023). The best of both worlds in network population games: Reaching consensus and convergence to equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Chen, K., Lim, J. Y., Kuan, K., & Soh, H. (2023). Latent emission-augmented perspective-taking (leapt) for human-robot interaction. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Zhang, B., & Soh, H. (2023). Large language models as zero-shot human models for human-robot interaction. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Taunyazov, T., Zhang, H., Eala, J. P., Zhao, N., & Soh, H. (2023). Refining 6-dof grasps with context-specific classifiers. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Fatir Ansari, A., Heng, A., Lim, A., & Soh, H. (2023). Neural continuous-discrete state space models for irregularly-sampled time series. In *International Conference on Machine Learning (ICML)*. [**Oral, < 2.4% of submissions**]

Balakrishnan, S., Bi, J., & Soh, H. (2022). SCALES: From Fairness Principles to Constrained Decision-Making. In *Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics and Society (AIES)*.

Chen, K., Fong, J., & Soh, H. (2022). MIRROR: Differentiable Deep Social Projection for Assistive Human-Robot Communication. In *Proceedings of Robotics: Science and Systems (R:SS)*.

Yam, K. C., Goh, E.-Y., Fehr, R., Lee, R., Soh, H., & Gray, K. (2022). When your boss is a robot: Workers are more spiteful to robot supervisors that seem more human. *Journal of Experimental Social Psychology*.

Hu, S., Leung, C.-W., fung Leung, H., & Soh, H. (2022). The dynamics of q-learning in population games: a physics-inspired continuity equation model. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*.

Ansari, A. F., Benidis, K., Kurle, R., Turkmen, A. C., Soh, H., Smola, A., ... Januschowski, T. (2021). Deep explicit duration switching models for time series. In *Neural Information Processing Systems (NeurIPS)*.

Taunyazov, T., Song, L. S., Lim, E., See, H. H., Lee, D., Tee, B. C., & Soh, H. (2021). Extended tactile perception: Vibration sensing through tools and grasped objects. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. [**Best Paper Award, 1 out of 2786 submissions**]

Ansari, A. F., Ang, M. L., & Soh, H. (2021). Refining deep generative models via discriminator gradient flow. In *International Conference on Learning Representations (ICLR)*.

Xie, Y., Zhou, F., & Soh, H. (2021). Embedding symbolic temporal knowledge into deep sequential models. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Chen, K., Lee, Y., & Soh, H. (2021). Multi-modal mutual information (mummi) training for robust self-supervised deep reinforcement learning. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Teh, N., Hu, S., & Soh, H. (2021). A theoretical framework for large-scale human-robot interaction with groups of learning agents. In *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*.

Balakrishnan, S., Nguyen, Q. P., Low, B. K. H., & Soh, H. (2020). Efficient exploration of reward functions in inverse reinforcement learning via bayesian optimization. *Advances in Neural Information Processing Systems*, *33*.

Ansari, A. F., Scarlett, J., & Soh, H. (2020). A characteristic function approach to deep implicit generative modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 7478–7487). [**Oral, < 5% of submissions**]

Kok, B. C., & Soh, H. (2020). Trust in robots: Challenges and opportunities. *Current Robotics Reports*, *1*(4), 297–309. doi: 10.1007/s43154-020-00029-y

Soh, H., Xie, Y., Chen, M., & Hsu, D. (2020). Multi-task trust transfer for human-robot interaction. *The International Journal of Robotics Research*, *39*(2-3), 233–249.

Lee, J., Fong, J., Kok, B. C., & Soh, H. (2020). Getting to know one another: Calibrating intent, capabilities and trust for human-robot collaboration. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*.

Gu, F., Sng, W., Taunyazov, T., & Soh, H. (2020). TactileSGNet: A Spiking Graph Neural Network for Event-based Tactile Object Recognition. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*.

Taunyazov, T., Chua, Y., Gao, R., Soh, H., & Wu, Y. (2020). Fast texture classification using tactile neural coding and spiking neural network. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*.

Bodala, I. P., Kok, B. C., Sng, W., & Soh, H. (2020). Modeling the interplay of trust and attention in hri: An autonomous vehicle study. In *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 145–147).

Lim, E., Kok, B. C., Wang, S., Lee, J., & Soh, H. (2020). Juiced and Ready to Predict Private Information in Deep Cooperative Reinforcement Learning. In *Companion of the ACM/IEEE International Conference on Human-Robot Interaction* (pp. 343–345).

Taunyazov, T., Sng, W., See, H. H., Lim, B., Kuan, J., Ansari, A. F., ... Soh, H. (2020). Event-driven visual-tactile sensing and learning for robots. In *Robotics: Science and Systems (RSS)*.

Yam, K. C., Bigman, Y. E., Tang, P. M., Ilies, R., De Cremer, D., Soh, H., & Gray, K. (2020). Robots at work: People prefer—and forgive—service robots with perceived feelings. *The Journal of Applied Psychology.*.

Tan, Z.-X., Soh, H., & Ong, D. (2020). Factorized inference in deep markov models for incomplete multimodal time series. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 34).

Ansari, A. F., & Soh, H. (2019). Hyperprior induced unsupervised disentanglement of latent representations. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, pp. 3175–3182). [**Spotlight**]

Ong, D., Soh, H., Zaki, J., & Goodman, N. (2019). Applying probabilistic programming to affective computing. *IEEE Transactions on Affective Computing*. doi: 10.1109/TAFFC.2019.2905211 [**Best Paper Award**]

Taunyazov, T., Koh, H. F., Wu, Y., Cai, C., & Soh, H. (2019). Towards effective tactile identification of textures using a hybrid touch approach. In *2019 International Conference on Robotics and Automation (ICRA)* (pp. 4269–4275).

Wu, G., Luo, K., Sanner, S., & Soh, H. (2019). Deep language-based critiquing for recommender systems. In *Proceedings of the 13th ACM Conference on Recommender Systems* (pp. 137–145).

Xie, Y., Bodala, I. P., Ong, D. C., Hsu, D., & Soh, H. (2019). Robot capability and intention in trust-based decisions across tasks. In *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 39–47).

Yaqi, X., Xu, Z., Meel, K. S., Kankanhalli, M., & Soh, H. (2019). Embedding symbolic knowledge into deep networks. In *Advances in Neural Information Processing Systems* (pp. 4235–4245).

Soh, H., Pan, S., Min, C., & Hsu, D. (2018, June). The transfer of human trust in robot capabilities across tasks. In *Proceedings of Robotics: Science and Systems*. Pittsburgh, Pennsylvania. doi: 10.15607/RSS.2018.XIV.033 [**Best Paper Finalist**]

Chen, M., Nikolaidis, S., Soh, H., Hsu, D., & Srinivasa, S. (2018). Planning with trust for human-robot collaboration. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 307–315). [**Best Paper Finalist**]

Kortschot, S. W., Sovilj, D., Jamieson, G. A., Sanner, S., Carrasco, C., & Soh, H. (2018). Measuring and mitigating the costs of attentional switches in active network monitoring for cybersecurity. *Human factors*, *60*(7), 962–977.

Lee, Y., Vo, T. V., Lim, K. W., & Soh, H. (2018). Z-transforms and its inference on partially observable point processes. In *International Joint Conference on Artificial Intelligence (IJCAI)* (pp. 2369–2375).

Sovilj, D., Sanner, S., Soh, H., & Li, H. (2018). Collaborative filtering with behavioral models. In *Proceedings of the 26th Conference on User Modeling, Adaptation and Personalization* (pp. 91–99).

Vo, T. V., & Soh, H. (2018). Generation meets recommendation: proposing novel items for groups of users. In *Proceedings of the 12th ACM Conference on Recommender Systems* (pp. 145–153). [**Best Paper Finalist**]

Azevedo, C. L., Deshmukh, N. M., Marimuthu, B., Oh, S., Marczuk, K., Soh, H., ... Ben-Akiva, M. E. (2017). Simmobility short-term: An integrated microscopic mobility simulator. *Transportation Research Record*, *2622*(1), 13–23.

Kinathil, S., Soh, H., & Sanner, S. (2017). Analytic decision analysis via symbolic dynamic programming for parameterized hybrid mdps. In *Twenty-Seventh International Conference on Automated Planning and Scheduling*.

Kortschot, S. W., Sovilj, D., Soh, H., Jamieson, G. A., Sanner, S., Carrasco, C., ... Langevin, S. (2017). An open source adaptive user interface for network monitoring. In *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 1535–1539).

Soh, H., Sanner, S., White, M., & Jamieson, G. (2017). Deep sequential recommendation for personalized adaptive user interfaces. In *Proceedings of the 22nd International Conference on Intelligent User Interfaces* (pp. 589–593).

Azevedo, C. L., Marczuk, K., Raveau, S., Soh, H., Adnan, M., Basak, K., ... others (2016). Microsimulation of demand and supply of autonomous mobility on demand. *Transportation Research Record*, *2564*(1), 21–30.

Soh, H. (2016). Distance-preserving probabilistic embeddings with side information: Variational bayesian multidimensional scaling gaussian process. In *International Joint Conference on Artificial Intelligence (IJCAI)* (pp. 2011–2018).

Soh, H., & Demiris, Y. (2015). Learning assistance by demonstration: Smart mobility with shared control and paired haptic controllers. *Journal of Human-Robot Interaction*, *4*(3), 76–100.

Soh, H., & Demiris, Y. (2014a). Incrementally learning objects by touch: Online discriminative and generative models for tactile-based recognition. *IEEE Transactions on Haptics*, *7*(4), 512–525.

Soh, H., & Demiris, Y. (2014b). Spatio-temporal learning with the online finite and infinite echo-state gaussian processes. *IEEE Transactions on Neural Networks and Learning Systems*, *26*(3), 522–536.

Soh, H., Su, Y., & Demiris, Y. (2012). Online spatio-temporal Gaussian process experts with application to tactile classification. In *Intelligent Robots and Systems (IROS), 2012 IEEE/RSJ International Conference on* (pp. 4489–4496). [**Best Cognitive Robotics Paper Finalist**]

Sarabia, M., Le Mau, T., Soh, H., Naruse, S., Poon, C., Liao, Z., ... Demiris, Y. (2013). iCharibot: Design and Field Trials of a Fundraising Robot. In *International Conference on Social Robotics* (pp. 412–421).

Su, Y., Wu, Y., Soh, H., Du, Z., & Demiris, Y. (2013). Enhanced kinematic model for dexterous manipulation with an underactuated hand. In *2013 IEEE/RSJ International Conference on Intelligent Robots and Systems* (pp. 2493–2499).

Soh, H., & Demiris, Y. (2011). Evolving policies for multi-reward partially observable Markov decision processes (MR-POMDPs). In *Proceedings of the 13th annual conference on Genetic and evolutionary computation* (pp. 713–720).

Zhang, T., Fu, X., Kwoh, C. K., Xiao, G., Wong, L., Ma, S., ... Lees, M. (2011). Temporal factors in school closure policy for mitigating the spread of influenza. *Journal of Public Health Policy*, *32*(2), 180–197.

Soh, H., Ong, Y.-S., Nguyen, Q. C., Nguyen, Q. H., Habibullah, M. S., Hung, T., & Kuo, J.-L. (2010). Discovering unique, low-energy pure water isomers: Memetic exploration, optimization, and landscape analysis. *IEEE Transactions on Evolutionary Computation*, *14*(3), 419–437.

Xiao, S., Xiao, G., Cheng, T., Ma, S., Fu, X., & Soh, H. (2010). Robustness of scale-free networks under rewiring operations. *EPL (Europhysics Letters)*, *89*(3), 38002.

Soh, H., Lim, S., Zhang, T., Fu, X., Lee, G. K. K., Hung, T. G. G., ... Wong, L. (2010). Weighted complex network analysis of travel routes on the singapore public transportation system. *Physica A: Statistical Mechanics and its Applications*, *389*(24), 5852–5863.

Zhang, T., Fu, X., Kwoh, C. K., Xiao, G., Wong, L., Ma, S., ... Lees, M. (2010). Effects of temporal factors in school closure policy for mitigating the spread of influenza. *Journal of Public Health Policy*.

Schreiber, M. J., Holmes, E. C., Ong, S. H., Soh, H. S., Liu, W., Tanner, L., ... others (2009). Genomic epidemiology of a dengue virus epidemic in urban singapore. *Journal of virology*, *83*(9), 4163–4173.

Nguyen, Q. C., Ong, Y. S., Soh, H., & Kuo, J.-L. (2008). Multiscale approach to explore the potential energy surface of water clusters. *The Journal of Physical Chemistry A*, *112*(28), 6257–6261.

Fu, X., Liew, C., Soh, H., Lee, G., Hung, T., & Ng, L.-C. (2007). Time-series infectious disease data analysis using svm and genetic algorithm. In *2007 IEEE Congress on Evolutionary Computation* (pp. 1276–1280).

Salahuddin, M., Hung, T., Soh, H., Sulaiman, E., Soon, O. Y., Sung, L. B., & Yunxia, R. (2007). Grid-based PSE for Engineering of Materials (GPEM). In *Seventh IEEE International Symposium on Cluster Computing and the Grid (CCGrid'07)* (pp. 309–316).

Soh, H., Ong, Y. S., Salahuddin, M., Hung, T., & Sung, L. B. (2007). Playing in the objective space: Coupled approximators for multi-objective optimization. In *2007 IEEE Symposium on Computational Intelligence in Multi-Criteria Decision-Making* (pp. 325–332).

Soh, H., Haque, S., Liao, W., Nadiminti, K., & Buyya, R. (2005). Gtpe: A thread programming environment for the grid. In *Proceedings of the 13th International Conference on Advanced Computing and Communications.*

Soh, H., & Kirley, M. (2006). mopga: Towards a new generation of multi-objective genetic algorithms. In *2006 IEEE International Conference on Evolutionary Computation* (pp. 1702–1709). [**Best Presentation Award**]

Recent Preprints and Reports

Wang, J., Chen, J., Liu, D., Song, W., Wang, Q., Ren, J., ... Soh, H. (2026). How to Better Train VLAs: Lessons Learned From the REAL-I Challenge at ICRA 2026. *arXiv preprint arXiv:2609.13679*. Retrieved from `https://arxiv.org/abs/2609.13679`

Shan, S., Zhou, C., Wang, R., Chen, X., Chen, X., Zhou, X., ... Yang, J. (2026). Learning Physical Interaction: A Survey of Tactile- and Force-aware Robot Learning. *arXiv preprint arXiv:2608.07558*. Retrieved from `https://arxiv.org/abs/2608.07558`

Marpally, S. R., Wang, A., Ghotavadekar, A., Ribeiro, R. A., Le, N., Bachiller-Burgos, P., ... Soh, H. (2026). ACME: A Multi-Cultural, Multi-Embodiment Social-Navigation Dataset. *arXiv preprint arXiv:2607.21964*. Retrieved from `https://arxiv.org/abs/2607.21964`

Bi, J., Wang, Q., Reddy, J., Lin, K., Khajikhanov, S., Gao, R., & Soh, H. (2026). Heterogeneous Tactile Transformer. *arXiv preprint arXiv:2606.29948*. Retrieved from `https://arxiv.org/abs/2606.29948`

Lee, Y.-C., Boonprakong, N., Tan, Y., Soh, H., Potanin, A., Kumar, V., ... Cristea, A. I. (2026). Reshaping Undergraduate Computer Science Education in the Generative AI Era. *arXiv preprint arXiv:2606.07545*. Retrieved from `https://arxiv.org/abs/2606.07545`

Wang, J., Chen, J., Liu, D., Ghotavadekar, A., Da, J., Kästner, L., & Soh, H. (2026). Change-Robust Online Spatial-Semantic Topological Mapping. *arXiv preprint arXiv:2605.02227*. Retrieved from `https://arxiv.org/abs/2605.02227`

Li, Q., Feng, L., Zhang, H., Li, W., Wang, L., Alahi, A., & Soh, H. (2026). Grounded World Model for Semantically Generalizable Planning. *arXiv preprint arXiv:2604.11751*. Retrieved from `https://arxiv.org/abs/2604.11751`

Zhang, B., Wang, M., & Soh, H. (2026). Not All Layers Need Tuning: Selective Layer Restoration Recovers Diversity. *arXiv preprint arXiv:2602.06665*. Retrieved from `https://arxiv.org/abs/2602.06665`

Soh, H., & Lim, E. (2026). Action Hallucination in Generative Vision-Language-Action Models. *arXiv preprint arXiv:2602.06339*. Retrieved from `https://arxiv.org/abs/2602.06339`

Zhao, Z., Niu, B., Soh, H., Hsu, D., & Lee, W. S. (2025). When Diffusion Breaks Constraints: Sequential Autoregressive Generation with RL and MCTS. *arXiv preprint arXiv:2512.01242*. Retrieved from `https://arxiv.org/abs/2512.01242`

Hagenow, M., Selvaggio, M., Yu, X., Wang, Y., Demiris, Y., Bobu, A., ... Shah, J. (2025). Shared Control/Autonomy: A Historical Perspective, Current Trends, and the Role of Generative AI. *TechRxiv preprint*. Retrieved from `https://wisc-rt2.github.io/shared_control_autonomy_survey/` doi: 10.36227/techrxiv.176617724.41163595/v1

PhD Students

Abdul Fatir Ansari (2017-2022, Joined Amazon AI as an Applied Scientist)
Dean's Research Excellence Award 2021, Research Achievement Award 2018
Xie Yaqi (2017-2023, joined CMU as a Research Fellow)
Research Achievement Award 2019
Sreejith Balakrishnan (2016-2023, co-supervised with Bryan Low, joined Resaro as a Senior Responsible AI Scientist),
Research Achievement Award 2021
Tasbolat Taunyazov (2018-2024, co-supervised with Yan Wu, A*STAR, joined Light Robotics),
School of Computing Best PhD Thesis Award in CS 2025; Dean's Research Excellence Award 2024; Research Achievement Award 2020
Chen Kaiqi (2019-2025, joined Light Robotics)
Research Achievement Award 2023, Research Achievement Award 2021
Bi Jianxin (2021-Present)
Shashank Rao Marpally (2021-Present)
Zhang Bowen (2021-Present)
Research Achievement Award 2024
Eugene Lim (2021–2026)
Alvin Heng (2021–2025)
Research Achievement Award 2023
Ce Hao (2024–2026)
Zhiwei Xue (2024-Present)
Jiaming Wang (2024-Present)
Oscar Jiang (2024-Present)

Postdoctoral Research Fellows

Linh Kastner (2023-Present)
Feng Zeyu (2020-2024, joined A*STAR)
Ze Gong (2022-2023, joined SMU as Research Fellow)
Siddharth Aravindan (2022-2023, joined Samsung as Scientist)
James Abraham (2021-2023, joined SMU as Research Fellow)
Hu Shuyue (2019-2021, now Research Scientist at Shanghai AI Labs)
Gu Fuqiang (2019-2020, now Professor at Chongqing University, China)
Lee Yong, (2019-2020, now Lecturer at Wuhan University of Technology)
Zhengkun Yi, (2018-2019, now Professor at Shenzhen Institute of Advanced Technology)
Indu Prasad (2018-2019, now Lecturer/Asst. Prof. at Southhampton Uni.)
Young Lee (2017-2018, joined Harvard University as a Postdoctoral Fellow)

Research Assistants

Sng Weicong (2018-2020, joined NUS as Graduate Tutor / PhD candidate)
Kok Bing Cai (2019-2020, Research Assistant, joined NUS Dept of Medicine)
Jethro Kuan (2020-2021, Research Assistant, joined ByteDance)
Zhu Changxi (2020-2021, joined Utrecht University as PhD candidate)
Tong Yichen (2021-2022, joined TUM as Masters Candidate)
Tan Yan Tui (2021-2022, joined Temasek Labs as Associate Scientist)
Haoyi Yang (2022, joined TUD as PhD Candidate)
Jiaming Wang (2022-2024)
Kelvin Lin (2022-Present)

PhD Examination & Thesis Committee

Qiang Wang (PhD, School of Electrical and Electronic Engineering, University College Dublin), External Examiner, 2024.
Min Jin Yang (PhD, Mechanical Engineering, KAIST), External Examiner, 2024 (Ongoing).
Dixant Mittal (PhD, Computer Science, NUS), Internal Examiner, 2024.
Bishwamittra Ghosh (PhD, Computer Science, NUS), Internal Examiner, 2023.
Ahswin Ram (PhD, Computer Science, NUS), Internal Examiner, 2023.
Chen Yizhou (PhD, Computer Science, NUS), Internal Examiner, 2022.

Siddharth Aravindan (PhD, Computer Science, NUS), Internal Examiner, 2022.
Ma Xiao (PhD, Computer Science, NUS), Internal Examiner, 2021.
Tian Meng (PhD, Mechanical Engineering, NUS), Inter-Faculty Examiner, 2021.

Academic Services

Program Committee: HRI'24 (co-Program Chair) HRI'22 (PC), HRI'21/23 (Technical Advances Theme Chair), HRI'20 (PC), IJCAI'20 (SPC), AAAI'21 (PC), AAAI'20 (PC), AAAI'19 (Senior PC), HRI'19 (LBR Chair), NeurIPS 2018-2021, IJCAI 2016-2018, IEEE SMC 2016.
Senior Associate Editor: ACM Transactions on Human-Robot Interaction (2025-present).
Associate Editor: ACM Transactions on Human-Robot Interaction (2021-2024), IEEE Robotics and Automation Letters (2021-2024), International Journal of Robotics Research (2023-present)
Ad-hoc Reviewer: JHRI, RSS 2019, HRI 2018, AAAI 2018, IEEE ICRA 2016-2022, IEEE IROS 2016-2021, ROMAN 17, ISRR 17, IEEE THMS, IEEE TNNSLS, Artificial Intelligence Journal, Autonomous Robots.

NUS Service

Department Ethics Committee	**Oct 2019 - Oct 2023**
Graduate Admissions Committee	**Oct 2018 - Present**
NUSAIL Associate Director	**2023 - 2024**

Affiliations

ACM, IEEE, AAAS

Grants (On-Going)

Amounts shown are total project funding in SGD.

Ministry of Education (MOE), AcRF Tier 2 **Sep 2026–Aug 2029**
Data-Efficient Expressive Policy Learning
PI SGD 805,129

NRF Investigatorship, National Research Foundation, Singapore
Learning Persistent Agent-Centric Representations for Embodied AI
PI **Jul 2026–Jul 2031**
SGD 2,944,660

National Research Foundation, Thematic Competitive Research Programme (T-CRP)
Bridging the Deployment Gap with Generative AI
PI **Feb 2026–Jan 2030**
SGD 3,788,335

Ministry of Education (MOE), AcRF Tier 1 **Jul 2025–Jun 2027**
Language-to-Behavior: Hierarchical Diffusion for Human Simulation
PI SGD 100,000

A*STAR **May 2025–Apr 2030**
Schaeffler-NTU Corporate Lab: Intelligent Mechatronics Hub
co-PI SGD 15,000,011

Grants (Completed)

Ministry of Education (MOE), DGAP Funding **Jan–Dec 2025**
ARENAGEN: Automated Generation of Realistic Simulation Environments Using Generative Models
PI SGD 50,050

Cornell-NUS Global Strategic Collaboration Award **2024**
Towards Real-World Robotic Caregiving: Bridging the Gap Between Roboticists and Healthcare Professionals
PI SGD 13,373

National Robotics Programme (NRP) **Oct 2023-Feb 2026**
Advancing Safe Human-Robot Interaction SGD 2,225,040
PI
co-PIs: Angela Yao, David Hsu, Yan Wu, Wei Yun

National Research Foundation (NRF) **Mar 2021-Mar 2026**
Centre for Advanced Robotics Technology Innovation (CARTIN) SGD 7,246,000
co-PI
PI: Xie Lihua (NTU) , co-PIs: Please see `https://www.ntu.edu.sg/cartin`

AI Singapore Programme, National Research Foundation (NRF) **Apr 2021-March 2025**
Trust to Train and Train to Trust: SGD 6,086,963
Agent Training Programs for Safety-Critical Environments
co-PI
PI: Pradeep Varakantham (SMU), co-PIs: Akshat Kumar (SMU), Arunesh Sinha (SMU), David Lo (SMU), Arvind Easwaran (NTU), Vivian Hsueh Hua Chen (NTU)

AI Singapore Programme, National Research Foundation (NRF) **Apr 2021-March 2025**
The "Other Me": Human-Centered AI Assistance In Situ SGD 9,551,520.00
co-PI
PI: David Hsu (NUS), co-PIs: Atreyi Kankanhalli (NUS), Lee Wee Sun (NUS), Angela Yao (NUS), Lee Gim Hee (NUS), Leong Tze-Yun (NUS), Desmond Ong (NUS), Zhao Shengdong (NUS), Pradeep Varakantham (SMU), Akshat Kumar (SMU), Dr. Basura Fernando (A*STAR), Lu Wei (SUTD), Malika Meghjani (SUTD), Georgios Piliouras (SUTD)

AI Singapore Programme, National Research Foundation (NRF) **Oct 2019-Sep 2022**
Assistive AI with Artificial Theories of Mind and Body SGD 750,000
PI

National Robotics Program (NRP), A*STAR **Oct 2019-Mar 2023**
Human Robot Interaction Phase 1 SGD 5,749,000
co-PI
PI: Haoyong Yu (NUS), co-PIs: Haizhou Li (NUS), David Hsu (NUS), Zhipin Lin (NTU)

National Robotics Program (NRP), A*STAR **June 2021-Aug 2022**
AiSkin 2a: Enhanced Multi-Sensorial Electronic Skins for Collaborative Robots SGD 1,799,508
co-PI
PI: Benjamin Tee (NUS), co-PIs: Zhuangjian Liu (A*STAR), Yan Wu (A*STAR)

National Robotics Program (NRP), A*STAR **Apr 2018-Mar 2021**
AiSkin: Next Generation Intelligent Robot Tactile Sensor SGD 1,717,000
co-PI
PI: Benjamin Tee (NUS), co-PIs: Zhuangjian Liu (A*STAR)

NUS, Office of the Deputy President (Research and Technology)
Startup Grant **Mar 2017-Mar 2019**
Intelligent Assistance for Collaborative Design with Deep Probabilistic Networks SGD 180,000
PI

Ministry of Education (MOE), Singapore **Oct 2017-Oct 2021**
Personalized Learning and Assistance for Sustainable Healthcare Management SGD 180,000
PI